HIKET — Materials and methods

complete version for review, to be merged into the second draft

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1 Materials and methods

The section first describes the data (the soil carbon observations, the litter input and the climate), then the six models and how they are started and calibrated, and then the calibrations and simulations built on that common setup: the comparison with a soil started in balance, the division of the modelled sink among its sources, and the simulations beyond the observations. The evaluation of the models against the observations closes the section.

1.1 Study plots and soil carbon observations

The study is based on data from permanent forest plots of the Finnish National Forest Inventory for which soil carbon has been measured in three campaigns: the eighth national forest inventory (VMI8), the BioSoil campaign of 2006, and the Komeetta campaign of 2024. After the exclusions described below, 456 plots are available for calibration, carrying 1205 plot-year observations of soil carbon in total. Of the 456 plots, 365 plots (80 per cent, carrying 959 observations) were used to fit the models, and 91 plots (20 per cent, carrying 246 observations) were held out and used only for evaluation. Plots were assigned to the two groups at random, and a plot was assigned as a whole, so that no plot contributes observations to both groups.

The first campaign is not a single year. Its plots were sampled region by region between 1986 and 1995: 69 plots in 1986, 76 in 1987, 103 in 1988, 54 in 1989 and 72 in 1995. Averaged over the plots, the sampling year of the first campaign is 1989, and that is the year at which the campaign is drawn whenever it is shown as a single point. In the calibration, each observation is compared with the model in the year in which that plot was actually sampled.

On the plots measured in all three campaigns (310 plots, the set on which every observed rate of change in this paper is computed), the mean stock of the groups sampled in 1986, 1987, 1988, 1989 and 1995 differs by 18.3 tonnes of carbon per hectare at first visit, which might reflect a trend within that same campaign. The five groups of plots defined by their first-campaign sampling year differ by 37.2 tonnes in 2006 and by 34.2 tonnes in 2024, years in which every plot of every group was measured in the same campaign. The difference between the five groups is therefore a property of which plots each group contains, and not of the year in which it was sampled, because the difference is larger when the sampling year is held constant.

[The exclusion rules (zero litter, absent organic layer, peat soils, constant litter series, whole-profile stock above 250 $\mathrm{t\,C\,ha^{-1}}$, and the 82 first-campaign observations without a recorded sampling year) are referred to as "described below" but are not yet written out; add one short paragraph listing them with the number of plots each removes.]

1.2 Soil carbon stocks on a single homogenised basis

Since the three campaigns were measured and processed under protocols that differed in ways that matter for a comparison between them, all three campaigns were rebuilt on a uniform basis, starting from the layer-by-layer carbon stocks measured in each of the three campaigns and applying one set of conventions to all three. Those stocks are corrected for the volume occupied by stones and for the bulk density of the fine earth. The volume of stones was measured in 2006 on every plot by the rod-penetration method of Viro (1952), in which a steel rod is driven into the mineral soil at the sampling points and the mean depth it reaches is converted into a stone volume, and the same stone volume is applied to the first campaign (Lehtonen et al., 2016; Kramarenko, 2012). The bulk density of the mineral layers was predicted from the equations of Tamminen and Starr (1994) for all layers of the first campaign and for most layers of 2006, and measured directly for part of the 2006 samples from the upper 20 cm (Kramarenko, 2012). [Confirm that the 2024 campaign uses the same stone volumes and the same bulk density equations.] The coarse-fragment correction matters because the coarse-fragment content of these plots has a median of 43 per cent by volume, so a mineral stock computed without it is too large by a factor of about 1.6.

The quantity used as the calibration target is the whole soil profile: the measured organic layer, plus the measured mineral soil to 40 cm, plus a modelled deep tail, integrated only as far down as the auger actually reached on that plot and in no case deeper than one metre. The depth to which the profile is integrated is therefore a per-plot quantity and not a fixed one metre, since using a fixed depth of, for example, one metre would invent soil carbon below bedrock on the 38 plots where augering stopped at 10, 20 or 40 cm. Below 40 cm the stock was extrapolated assuming that the carbon content decreases exponentially with depth. One rate of decrease was fitted for each soil class of the national soil map, jointly on the profiles of all three campaigns, and the level of each profile's curve was set by that profile's own measured mineral layers. The first campaign was sampled in different depth intervals (0–5, 5–20 and 20–40 cm, against 0–10, 10–20 and 20–40 cm in the two later campaigns); the fit uses the actual interval of each layer, so the two layer schemes need no conversion. The depth reached by the auger is recorded only in 2006 and, like the stone volume, is applied to the first campaign. In the first campaign the carbon content of the mineral layers below 5 cm was mostly predicted from the loss of mass on ignition rather than measured (Kramarenko, 2012), and because the deep tail is scaled from those layers, the deep tail carries the same uncertainty. The extrapolated part is about 15 per cent of the profile stock in each of the three campaigns. The modelled deep tail was validated against the measured 40–80 cm layer of the BioSoil campaign, which was held out of the fit: the median ratio of predicted to observed carbon in that layer is 0.96, with a bias of -2.5 tonnes of carbon per hectare.

The rebuilt target was checked against the national figures published for the two campaigns for which such figures exist. Applying the same plot exclusions as the official calculation, and integrating over the same layers, the rebuilt target returns weighted national means of 59.06 and 61.05 tonnes of carbon per hectare for 2006 and 2024 against official values of 59.06 and 61.05, that is, agreement to the nearest 0.01 tonnes per hectare on 446 plots.

Two further steps were taken with the first campaign, both of them established from the structure of the measurements themselves rather than from the fit. First, the layers of the three campaigns were standardised, because they were recorded differently: the organic layer of the first campaign contains the fermentation and humus horizons only, whereas the organic layer of the two later campaigns also contains the litter and moss layer. Each plot's own litter and moss layer from 2006

was therefore added to its first-campaign organic layer (for the 20 calibrated plots without a value from 2006, the median of all plots), so that the same compartment is compared across the three campaigns. Second, the first campaign is treated as what it is, a campaign whose fieldwork ran over several years rather than a campaign carried out in a single year. The year in which each plot was actually sampled is on record, and those years, rather than one nominal year for the whole campaign, are the years at which the models are compared with the observations.

On this final basis, over the 1205 observations used in this study, the median whole-profile stock is 61.4 tonnes of carbon per hectare for the first campaign, 65.3 for 2006 and 66.8 for 2024, and the observations vary between plots with a coefficient of variation of 0.36 to 0.48 depending on the campaign. [Supplementary figure: F2_baseline]

[Data availability and acknowledgement: the soil carbon measurements are not a published dataset, so the statement has to say where they come from, who compiled them and what of the compiled form can be deposited. Settle this with the people who hold them before submission.]

1.3 Litter input

Litter input is taken from the annual reconstruction of tree biomass and litter input for the permanent plots of the Finnish National Forest Inventory by Ťupek et al. (2026), published as an open dataset. In that reconstruction, the tree biomass measured by the inventory is interpolated to every year with the help of satellite-derived basal area, and litter is computed as the biomass of each tree component multiplied by a turnover rate for that component, plus the branches, foliage, stumps and roots left on site when a drop in basal area identifies a harvest; the full method is given in Ťupek et al. (2026). The product supplies, for each plot and year, the carbon in litter from trees, separated into non-woody litter (foliage and fine roots), fine woody litter (branches and coarse roots) and coarse woody litter (stumps), and each of the three classes is further divided into the four chemical fractions used by the Yasso models, using proportions specific to each tree species and component. The product does not contain a separate term for trees that die without being harvested, and it does not include the understorey vegetation, because the product is a tree litter product by design. Across the study plots the mean litter input carried by the product is $\bar{J} = 2.511$ tonnes of carbon per hectare per year.

The national greenhouse gas inventory estimates the same flux for the mineral forest soils of southern and northern Finland, from the forest inventory and biomass models (Statistics Finland, 2025). On the part that the two estimates have in common, the litter from the turnover of living trees and from harvest residues, they agree in level in southern Finland, where most of the study plots lie, and neither rises after about 2006; in northern Finland the inventory estimate lies higher and keeps rising [Supplementary figure: S18_litter_vs_inventory]. The inventory estimate also carries the litter of trees that die without being harvested, a term that grows over the period and that the plot product does not contain. [confirm with LUKE that the two regions of the inventory data match the region code of the study plots]

The first year of the product is treated separately. The median litter input of 1985 is 0.060 tonnes of carbon per hectare per year against 1.398 in 1986, a step that the series does not repeat in any later year. Because 1985 is both the first year of the simulation and the year in which the first soil observations fall, the value for 1985 was reconstructed for each plot and each chemical fraction by fitting a linear trend over 1986–1990 and extrapolating one year backwards. The reconstruction places 1985 just below 1986, which is the direction the growing-stock record requires.

1.4 Climate

Climate forcing comes from the gridded daily weather observations of the Finnish Meteorological Institute, interpolated from the station network onto a 10 by 10 km grid by kriging with external drift (Aalto et al., 2016) and extended by the Institute to the present. The grids were obtained through the internal weather database of the Natural Resources Institute Finland, which mirrors them from the Finnish Meteorological Institute. From the grid cell nearest to each plot we took the daily mean, minimum and maximum temperature, precipitation and global radiation for 1961–2025, and each model takes from those series the variables it requires. The record is used in two ways. From 1985 onwards each year of the simulation is driven by the climate observed in that year. Before 1985 the simulation is driven by one constant climate, the mean of 1961–1984 at each plot (a plot mean of 2.2 degrees Celsius, against 2.7 for 1985–2004), so that the soil enters 1985 adjusted to the climate it had lived in; the same long-term mean serves as the reference climate wherever one is needed later (Sections 1.7, 1.13 and 1.14).

Because the response of decomposition to climate is not a straight line, the average of the response over the years is not the response evaluated at the average climate. Averaging the climate first and evaluating the response afterwards biases the climate modifier by 5 to 15 per cent in boreal conditions. The climate modifier is therefore computed for each year and averaged afterwards, in all six models.

1.5 The six models

Six models are used (three Yasso versions, plus three simpler benchmark models), and the two groups have different roles. Yasso is the model this study is focused on, because Yasso is the model the Finnish greenhouse gas inventory is currently based upon. All three published versions of Yasso are included: Yasso07, Yasso15 and Yasso20. Alongside the three versions of Yasso we included three simple models with one, two and three carbon pools, denoted SP1, TP2 and TP3. The three simple models are there as a benchmark, since their behaviour is transparent enough so that any result obtained with Yasso can be checked against a structure simple enough to reason about quickly.

Due to their different structures and designs, the number of free parameters differs greatly across the six models, from six free parameters in SP1 to twenty-six in Yasso15 and Yasso20. The free parameters of the three versions of Yasso are the twelve fractions describing transfers between pools, the climate parameters, the parameters describing the effect of the size of woody litter, and the two input parameters described in Section 1.6.

The decomposition rates of Yasso are not fitted, and are held at their published values. Holding the rates fixed is a deliberate choice to keep the model anchored to its original definitions. Since decomposition rates and transfer fractions do not act independently on the carbon stock but are always interacting, a slower rate and a larger transfer into a slow pool produce very similar trajectories, so fitting both together adds little information and makes the result harder to read.

Whether the pools of a soil carbon model can be identified with fractions of organic matter isolated in the laboratory is a long-standing question in soil carbon modelling, and has been examined at length (von Lützow et al., 2007). A chemical fractionation is itself a discretization: it divides a continuum into a small number of classes, and where the boundaries between classes fall is set by the method chosen, so that the fractions are defined by the procedure that isolates them (von Lützow et al., 2007). The material being divided is not necessarily a set of discrete compounds,

since the chemical diversity of soil organic matter is large enough that the diversity can be treated as a continuum (Lehmann and Kleber, 2015). Yasso was built on the assumption that the pools can be identified chemically, and that litter consists of groups of organic compounds, each group decomposing at its own rate regardless of the litter it came from, and defined its decomposition compartments accordingly as extractives, celluloses and lignin-like compounds, followed by two compartments of humus (Liski et al., 2005). All following Yasso versions are built on the same assumption, and the models have been calibrated on data derived from such assumption, which is therefore a fundamental part of the model. The same aim, of matching the pools of a model with fractions that can be isolated in the laboratory, has been pursued for other models, for example through fractionation schemes designed to correspond to the pools of RothC (Zimmermann et al., 2007). Whatever the pools are meant to represent, however, and whatever the nature of soil organic matter, which remains debated, a compartmental model defines each pool by its decomposition rate: the rate groups carbon by how long it stays in that pool, its turnover time (Sierra et al., 2017), and changing a rate changes what the pool is.

We therefore treated the rates as part of the definition of each model, kept the rates as the published parameterisation set them, and calibrated the transfer fractions. In all three versions of Yasso the decomposition rates, together with the fraction of decomposed carbon that enters the humus pool, are fixed at their published values: they are constants of the model and not parameters of the calibration, and no prior is placed on them.

1.6 The uncertainty of the litter input

The litter input is among the least certain of the quantities that drive a soil carbon model. Litter is not measured on the plots but derived from tree biomass. Parts of the input, such as the litter of the understorey, the litter of trees that die naturally and root exudates, are missing from the product used here (Section 1.3), and the input carries year-to-year variation that periodic inventories cannot capture. Uncertainty analyses of the Finnish soil carbon estimate have already identified the litter input, together with the initial state of the soil, as the dominant sources of uncertainty (Peltoniemi et al., 2006; Lehtonen and Heikkinen, 2016). Those analyses propagated the uncertainty forward by Monte Carlo simulation, sampling the input, and in Lehtonen and Heikkinen (2016) also the parameters of Yasso07, from error distributions derived from the data sources, and running the soil model with each sample. We instead estimate the error of the input from the soil observations themselves, together with the parameters of the models, so that the correction applied to the input and the uncertainty of that correction are both learned from the data and carried into every result.

The correction is a single multiplier, σ_{input}, the *input multiplier*, applied to the litter input of the product over the whole simulation. The prior of σ_{input} is centred on 1.08, the ratio of the total litter input to Finnish upland soils estimated by Lehtonen and Heikkinen (2016), which includes the understorey, to the tree litter input of the product used here, 2.70 against 2.511 tonnes of carbon per hectare per year. The prior is set on the litter input that σ_{input} produces, which is kept within 0.05 to 8.7 tonnes of carbon per hectare per year, the upper end of the net primary production reported for evergreen forests (Gower et al., 2001), so that no value of σ_{input} can produce an impossible input. Within that range the prior gives σ_{input} a median of 1.07 and a 95 per cent probability of lying between 0.75 and 1.45. Because the multiplier is one number for the whole country, it rescales only the national level of the input and not the input of individual plots, whose errors are carried by the error model described below.

The same reasoning is extended to the initial state. The litter input of 1917, which sets the state of the soil at the start of the simulation, is not known either, and is estimated through a second parameter, σ_{init}, the *initial input ratio*, the ratio of the litter input of 1917 to the litter input of 1985. The prior of σ_{init} is centred on 0.90, the ratio of the growing stock of Finnish forests in 1917 to that in 1985 (Korhonen et al., 2024) converted to a ratio of litter input with the relation between litter input and growing stock observed in the product, and is constructed in the same way, on the litter input of 1917, giving a median of 0.93 and a 95 per cent range of 0.58 to 1.49; the reconstruction of Liski et al. (2006), based on the same growing-stock record, implies a similar ratio, about 0.86. Both parameters are estimated together with the model parameters, so that the uncertainty of the input and of the starting state propagates into the initial state, into the simulated trajectory and into the predictive intervals of every model.

1.7 Transient initialisation

Rather than starting each plot from a soil in balance with its present litter input, each draw of the parameter vector starts the soil in 1917 and runs it forward to the first soil observation before the comparison with data begins. The year 1917 is a convention, chosen as the year of Finnish independence. The state of the soil in 1917 is obtained conventionally, with a steady state assumption with a constant litter input equal to the estimated litter input of 1917 and constant climate (calculated as the mean of 1961–1984, Section 1.4). The litter input is then raised along a prescribed shape over the sixty-eight years from 1917 to 1985, still under the same constant climate, after which the simulation is driven by the annual litter and climate of the record. A constant climate is used before 1985 because the records start in 1961 and do not reach back to 1917; the mean of the earliest observed decades is the closest available measure of the climate in which the soil developed before the observations began.

The shape along which the litter input is raised is taken from the reconstruction of the litter input to Finland's forest soils published by Liski et al. (2006), who estimated that input annually from 1922 onwards by combining national forest inventory data with biomass and turnover models.

Three decisions were made about that reconstruction.

First, only the tree-derived terms are used, meaning tree litter together with harvest residues (and natural tree mortality), because those are the terms closest to the litter product used after 1985. The plot-level product of Ťupek et al. (2026) has no term for natural mortality: a tree that dies is registered only as a loss of basal area, and its stem does not enter the soil. Natural mortality is, however, only 2 to 4 per cent of the tree-derived input in Liski et al. (2006), and leaving it out of the shape changes no year by more than 0.04 on the scale of zero to one defined below. The ground vegetation from Liski et al. (2006) is left out for uniformity with the product used after 1985, assuming that it is absorbed by the input multiplier described in Section 1.6. Including it before 1985 but not after would make that multiplier mean two different things on either side of the join.

Second, the national totals are converted to a per-hectare basis by dividing by the area of upland forest soil, derived from Liski et al. (2006). Liski et al. (2006) report the carbon of upland soils both as a national total and per unit area, and the ratio of the two gives an upland soil area of 13.90 million hectares in 1922 and 15.22 million hectares in 2004. The area is held at the first value until 1965, increases linearly to the second over 1965–1980, the period in which Korhonen et al. (2024) locate most of the expansion of productive forest land, and is held constant afterwards.

Third, only the shape is taken from Liski et al. (2006) and not the level: the series is rescaled to run from zero in 1917 to one in 1985, so that the reconstruction supplies only the path between the two endpoints. The litter input in year t of the pre-run is $J(t) = J_{1917} + (J_{1985} - J_{1917}) \times s(t)$, where $s(t)$ is the rescaled shape; the same weighting is applied to each chemical fraction and size class of the input, so that the composition of the input is that of the plot in 1985–1989 throughout and only its amount changes. The two endpoints (J_{1917} and J_{1985}) are set by the two input parameters described in Section 1.6: the input at the end of the path is the litter input of the product, averaged over 1985–1989 for each plot and scaled by σ_{input}, and the input at the start of the path is a fraction σ_{init} of the input at the end.

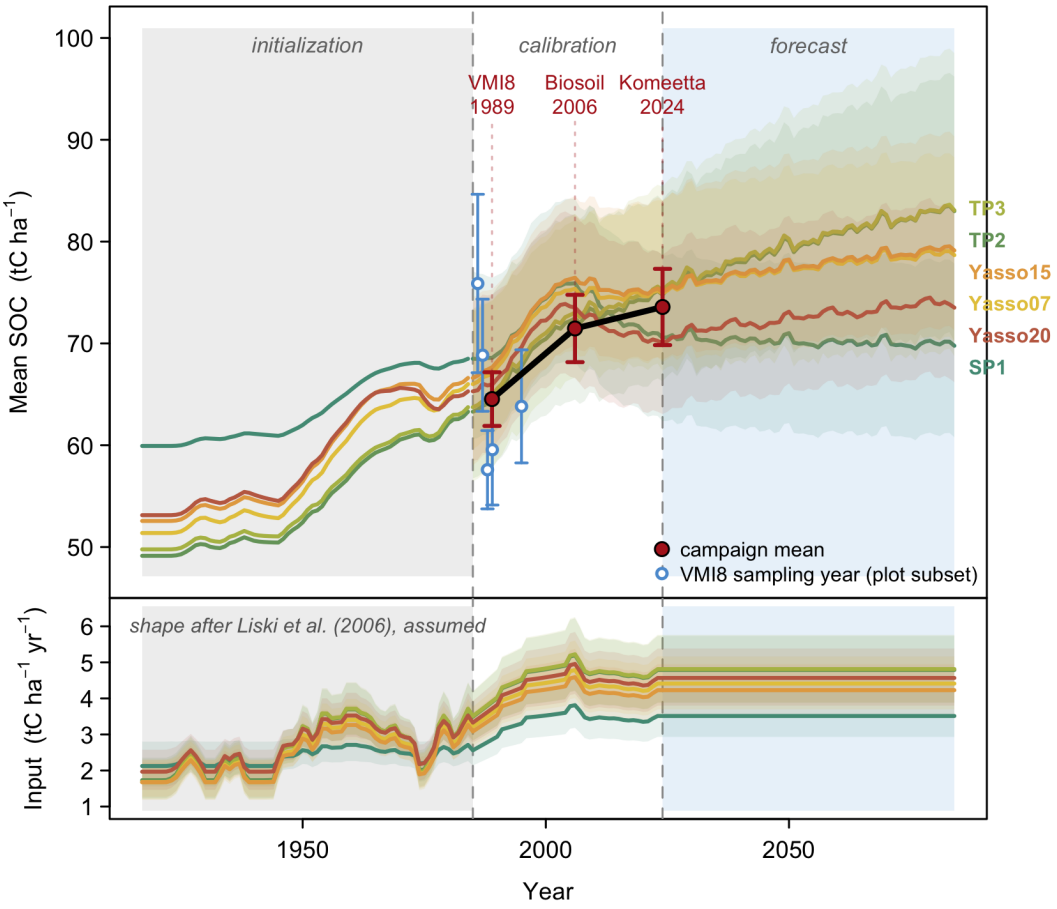


Figure 1: How the starting state of the soil is obtained. Rather than starting each plot from a soil already in balance with its present litter input, the simulation starts in 1917 from a soil in balance with the litter input of 1917 and the mean climate of 1961–1984, and the litter input is then raised over the following sixty-eight years along a shape taken from the reconstruction of Finland's forest litter input by Liski et al. (2006), after which the simulation is driven by the annual litter and climate of the record. The litter input of 1917 is not known and is estimated as part of the calibration.

1.8 Keeping the models as comparable as possible

The six models cannot be given the same amount of external information. The models differ in structure and in the number of their parameters, and the published literature provides different information for each of them: the three versions of Yasso come with their own calibrations, whereas the three simple models have no calibration for forest soils. The more the models differ in the information they are given, in their assumptions and in their freedom, the less a difference between their results can be attributed to their structure. We therefore reduced that difference as far as the models allow, through one compromise for each class of parameter, summarised in Table 1. The compromises reduce the asymmetry between the models without removing it, and this must be considered when interpreting the results.

1.8.1 Decomposition rates

The rates of the three versions of Yasso are fixed at their published values, for the reasons given in Section 1.5. The three simple models have no rates of their own for forest soils, and take theirs from the ICBM model (Andrén and Kätterer, 1997): the fast rate is fixed, and the slow rate or rates are given a very informative prior centred on the ICBM value, so that the data can move them only slightly. The slow rates are pinned rather than fixed because the ICBM values cannot be transferred to a forest soil unchanged. ICBM was calibrated on an arable long-term experiment, and cropland and forest soils differ in what controls decomposition. Moreover, a rate is only meaningful together with the climate response it was estimated with, and the simple models combine the rates of ICBM with the climate response of Yasso07; the rates of ICBM were divided once by the value of the Yasso07 climate modifier calculated on the climate of the experiment on which ICBM was calibrated (0.94). Fitting the rates freely (or under wide priors), on the other hand, would give the simple models a freedom that the versions of Yasso do not have, and a narrow prior around the ICBM value is the compromise between the two.

1.8.2 Transfer fractions

The fractions that describe how carbon passes between pools are free in every model that has them: twelve in each version of Yasso, and the fraction of decomposed carbon passed on to the slower pool in the simple models, of which SP1 has none, TP2 one and TP3 two. All fractions carry a prior of the same width, 0.4 on the logit scale, centred on the published value of each model, which for the simple models is the humification coefficient of ICBM.

1.8.3 Climate response and size of woody litter

The three versions of Yasso take the centres and widths of their climate priors from their own calibrations: Yasso07 from Tuomi et al. (2009, 2011); Yasso20 from the sample of 10 000 parameter sets of Viskari et al. (2022), distributed with the model code (YASSOmodel, 2022); and Yasso15, whose calibration has not been published in a journal, from the posterior sample distributed with its model code (YASSOmodel, 2020). The simple models, which have no published climate response of their own, borrow the climate response of Yasso07, centre and width. Only the versions of Yasso describe how decomposition depends on the size of woody litter, with priors taken from their own calibrations; the simple models have no such term.

1.8.4 The two input parameters

Every model carries the two parameters described in Section 1.6, the input multiplier σ_{input} and the initial input ratio σ_{init}, with the same prior in every model.

Table 1: How the prior of each class of parameter was set.

[Parameter-class by prior-criteria table: for each class (decomposition rates, transfer fractions, climate response, size of woody litter, the two input parameters), the source of the centre, the source of the width, the transformation and constraint, and which models carry it. The worked version is in `Prior_specs/PRIOR_HOMOGENIZATION_PLAN.md`, section 2.]

1.9 Numerical integration

Each model is run forward one year at a time, from 1917 to the last observation, with the litter input and the climate of each year held constant within that year. Within each year the equations of every model are solved exactly rather than stepped forward. For the one-, two- and three-pool models, in which carbon moves in one direction only, the exact solution has a closed form, an exponential approach of each pool to its balance value, and is evaluated directly. In the three versions of Yasso carbon moves between pools in every direction, and the solution, the exponential of the matrix of rates and transfers, is computed in the Fortran code of the models by scaling and squaring (Moler and Van Loan, 2003). All models are run with an annual time step, the step for which Yasso was designed and calibrated; the seasonal course of temperature enters through the annual temperature amplitude in the climate modifier (Tuomi et al., 2009), which the simple models share.

1.10 Mean transit time as a common measure of a model

The six models differ in the number of their pools and in how carbon moves between those pools, so their parameters cannot be compared directly. The models are compared instead through one number, the mean time a unit of carbon spends in the soil between entering it as litter and leaving it as carbon dioxide, which Sierra et al. (2017) call the mean transit time. The same quantity is often called the mean residence time; Sierra and coauthors show that the term residence time has been used for at least three different quantities, and we follow their convention. For a model whose pools lose carbon in proportion to the carbon they hold, and whose rates do not change with time, the mean transit time equals the carbon the system holds at steady state divided by the carbon entering it (Rasmussen et al., 2016; Sierra et al., 2017). Each model is therefore given an input of one tonne of carbon per hectare per year, and its steady-state stock is computed directly: as a formula for the three simple models, and as the solution of one linear system for the three versions of Yasso. Because the models are linear, the ratio of stock to input does not depend on the input chosen.

The mean transit time computed in this way is a property of the model alone. Neither σ_{input}, nor σ_{init}, nor any observation enters the calculation, so different models (and the calibrated and the published parameterisations of the same model) are compared on the same scale. Because the models respond to climate and to the quality of the litter, the mean transit time for every model and every parameter set is evaluated at one reference condition: the mean climate of the study plots over all the observed years (mean annual temperature 3.4 °C, temperature amplitude 13.8 °C, annual precipitation 591 mm), and the mean litter of the study plots, divided into 0.766 non-woody,

0.219 fine woody and 0.014 coarse woody litter, each with its own chemical composition. The values reported are therefore those of each model under that condition, not those of the real soil under changing conditions; the comparison between parameterisations does not depend on the reference, because both sides are evaluated at the same one.

1.11 The error model

Observations are compared with predictions on the logarithmic scale, so that the error is proportional to the size of the stock. The total standard deviation of the error is fixed at 0.800 on that scale, about twice the spread of the observed stocks around the national mean of each campaign, 0.39. That spread is the error of an idealised model that follows the national mean exactly and knows nothing about individual plots. A process-based model adds information that the measurements do not contain, in the litter input, the climate and the representation of decomposition, and is expected to depart from the measurements where the measurements are in error. The realistic error is larger still, for four reasons: the litter input carries errors of its own at each plot, which counted as an independent error alone raise the total to 0.57; the observed errors have heavier tails than a normal distribution; outlying plots were excluded from the sample; and part of the measurement error, in the stone volume, the bulk density and the modelled deep tail of the profile, is shared between plots and does not appear in a spread between plots. A value at the upper end of the plausible range was chosen, because a larger error weakens the pull of the observations against the priors and keeps conclusions drawn from the observations conservative.

Observations of soil carbon are not independent of one another: observations of the same plot, of the same part of the country and of the same campaign share whatever the models get wrong about that plot, that region or that protocol. The error of each observation is therefore divided into four parts,

$$\log y_{ij} \;=\; \log f_{ij}(\theta) \;+\; u^{R}_{r(i)} \;+\; u^{P}_{i} \;+\; u^{C}_{j} \;+\; e_{ij},$$

where y_{ij} is the stock observed on plot i in campaign j, $f_{ij}(\theta)$ the stock predicted by the model, u^R a part shared by all plots in the same band of latitude, u^P a part shared by all observations of the same plot, u^C a part shared by all observations of the same campaign, and e a part that belongs to the observation alone. Each part has a mean of zero and a fixed standard deviation, and no part is estimated for any plot, band or campaign: the likelihood is averaged over all the values the shared parts could take, which for normal errors can be done exactly and leaves only a correlation between observations that share a plot, a band or a campaign.

The standard deviations of the shared parts were taken from the observations after removing the mean of each campaign: 0.117 for the regional part, the spread between the means of eight bands of latitude holding equal numbers of plots, and 0.396 for the plot part, from how strongly two observations of the same plot vary together. The part that belongs to each observation alone, 0.685, is what remains of the total once the shared parts are accounted for, the variances (the squares of the standard deviations) of the four parts adding up to 0.800^2. Dividing the error does not change how far a single observation may lie from the model, but observations that share part of their error partly repeat one another, and the 1205 observations carry about as much information on the national mean as 220 independent observations would. Both published calibrations of Yasso treated the observations as independent (Tuomi et al., 2009; Viskari et al., 2022), the second naming the correlation of errors within a time series as an effect still to be included, and treating correlated

observations as independent makes parameter uncertainties too narrow (Oberpriller et al., 2021).

The standard deviations of the campaign parts cannot be measured, because a shared offset between campaigns and a genuine change of the stock look the same in the observations, and were therefore prescribed: 0.030 for each of the two later campaigns, which followed similar protocols, and 0.060 for the first, which differs from them in three documented ways. Its organic layer excluded the litter and moss layer, added here from each plot's measurement of 2006; its fieldwork extended over a decade; and the carbon content of its mineral layers below 5 cm was mostly predicted from the loss of mass on ignition rather than measured (Kramarenko, 2012).

1.12 Sampling

Each model was calibrated with five independent runs of 50 000 iterations, of which the first 5 000 were discarded, with the differential evolution Markov chain sampler of ter Braak and Vrugt (2008), which adapts its proposals to strongly correlated posteriors, as implemented in the `BayesianTools` package (Hartig et al., 2023), version 0.1.9, under R 4.6.1 (R Core Team, 2023). Each run maintains three interacting chains, so that the five runs together provide fifteen chains, and every calibration in this paper uses the same settings. Convergence was assessed with the potential scale reduction factor and the effective sample size: over all eighteen calibrations (six models, each with the transient initialisation, with a balanced start, and with a balanced start and the climate held) the largest potential scale reduction factor of any parameter was 1.009 and the smallest effective sample size 1716.

The information that the observations add to each parameter was measured as the Kullback–Leibler divergence of the marginal posterior from the marginal prior, estimated from kernel densities of the samples and expressed in nats. The divergence is zero when the observations say nothing about a parameter; for scale, a posterior with the centre of its prior and a quarter of its width gives about 0.9 nats, and a posterior with the width of its prior moved by one prior standard deviation gives 0.5 nats.

1.13 The comparison with a soil started in balance

To test whether the transient initialisation is needed, every model was calibrated a second time with the soil started in balance, identical in data, error model, priors and sampling, and different in one factor only: in 1985 the soil is set to the state at which its gains and losses balance under the litter input of 1985–1989, scaled by σ_{input}, and the reference climate of 1961–1984. The initial input ratio σ_{init} is fixed at one, which makes the litter path of Section 1.7 flat, and is removed from the parameters; the prior of σ_{input} is unchanged. The balanced start is the steady-state assumption applied at the start of the measurements, and not the procedure of any particular inventory, and is the strongest fair version of that assumption: every other parameter is recalibrated, so the models are free to compensate for the missing deficit through the litter input, the transfer fractions and the response to climate. A soil started in balance is not expected to stay flat, because the litter input rose by about half until 2006; what distinguishes the two starts is the size of the accumulation they produce. The expectations for the comparison were written down before the calibrations were run (Supplement [S?]). The two calibrations are compared on the trajectory of the national mean against the three campaigns, on the rates of change over 1985–2006, 2006–2024 and 1985–2024, on the position of each model along the trade-off between litter input and mean transit time, and on the

fit. The comparison rests mainly on the rates rather than on the log-likelihoods, which are directly comparable here but inherit the overconfidence that the error model only partly removes.

1.14 The partitioning of the modelled sink

The sink each calibrated model produces over 1985–2024 was divided among three sources: the state inherited from before 1985, the change of the litter input since 1985, and the climate. One further calibration was made for every model, with the soil started in balance *and* the climate held at its 1961–1984 reference in every year after 1985, in the calibration as in the simulation. Each model is thereby calibrated three times on the same measurements, adding one assumption at a time:

1. the transient initialisation, with the observed litter and climate;
2. a balanced start, with the observed litter and climate (Section 1.13);
3. a balanced start, with the observed litter and the climate held at the reference.

With S_1, S_2 and S_3 the sinks of the three calibrations, the division is

$$S_1 \;=\; \underbrace{(S_1 - S_2)}_{\text{inherited state}} \;+\; \underbrace{(S_2 - S_3)}_{\text{climate}} \;+\; \underbrace{S_3}_{\text{litter change}} \quad ,$$

summarised in Table 2; the three parts add up to the sink of the first calibration by construction.

Table 2: The three parts of the modelled sink. Each part is the difference between two calibrations fitted to the same measurements, which differ in one assumption.

part	calibrations	the assumption that differs	what the part measures
inherited state	$1 - 2$	starting state: estimated, or balanced in 1985	what the deficit inherited from before 1985 adds to the sink
climate	$2 - 3$	climate after 1985: observed, or held at the 1961–1984 reference	what the observed climate adds, acting on the litter that actually arrived
litter change	3	—	the sink of a soil started in balance under an unchanging climate, which exists only because the litter departs from its 1985–1989 level

Each part therefore compares two fitted models that differ by one assumption, including the change of the other parameters that the data demand once the assumption is made, and is not the effect of one source on a model held fixed. The joint effects of climate with the inherited deficit and with the litter change are assigned by the order of the calibrations, to the inherited and to the climate part respectively. The parts are computed for the plots measured in all three campaigns (310 plots) and 50 posterior draws of each calibration, and their additivity and a zero sink under constant litter and climate were checked for every draw. The size of the climate part depends on the reference climate, whereas the ordering of the parts does not. [Five terms (two further calibrations) planned with the new input data.]

1.15 Simulations beyond the observations

The calibrated models were then run beyond the observations. The simulations describe the calibrated models on the study plots, per hectare, under stated forcing, and are not a forecast of the national soil carbon balance. Each model is run for 60 years after 2024 with the litter input held at its 2024 value and the climate of the last 20 observed years repeated three times, so that the projection contains no change of climate or of forest growth; rates over the projection are computed over whole 20-year cycles, because the repeated climate alternates between a cooler and a warmer decade. The projection uses 100 posterior draws and the plots measured in all three campaigns, for both the transient and the balanced-start calibration. The projection is the basis of three further quantities:

The carbon remaining to be accumulated before balance. The equilibrium stock of each plot is the stock at which gains and losses would balance under the litter and climate of the projection, computed as for the mean transit time (Section 1.10); the remaining accumulation is the equilibrium stock divided by the stock of 2024, minus one. The remaining accumulation is a quantity of carbon, not a time: how long the soil would take to reach balance is set by its slowest pool and is not computed here.

What could happen with more litter input. The projection is repeated with the litter input increased by 20 per cent from 2025. Because the models are linear in their input, the sink over any period is a straight line in the input multiplier, passing through the projections with the input held and increased by 20 per cent; the increase of input that would hold the observed rates of 1985–2024 and of 2006–2024 over the first 10 and 20 years of the projection is read off that line, without further simulation. Without further warming, these increases are lower bounds.

What could happen with warming. A linear rise of temperature, from zero in 2025 to its full value in 2084, is added to the repeated climate, and the stock in 2084 is compared with that of the projection without the rise. The rises are the warming still to come under SSP1-2.6, SSP2-4.5 and SSP5-8.5 for Finland (Ruosteenoja and Jylhä, 2021): the published warming relative to 1981–2010, minus the 1.0 °C by which the repeated climate of 2005–2024 already exceeds that of 1985–2010 on the study plots, giving 1.3, 2.6 and 5.4 °C. The litter input is held constant at its 2024 value, with no response of forest growth, so the result isolates the response of decomposition to warming, and it is a sensitivity, not a projection. The response is computed for three parameterisations of each version of Yasso, each started from its own state under the pre-run climate: the transient and the balanced-start calibrations (40 posterior draws each), and the published parameterisation, from draws of the published posterior (40 posterior draws each) for Yasso15 and Yasso20 and from the published values for Yasso07. None of the published kinetic parameters is refitted. The published posterior samples carry no input parameters for our plots, so both are supplied: the initial input ratio σ_{init} is taken as the posterior median of the transient calibration, and the input multiplier σ_{input} is chosen so that the published model fits the three observed campaign means as closely as possible, fitted for the draw closest to the published median transit time and, for the other draws, scaled by the ratio of their transit times to that of the fitted draw. Because the stock is proportional to the input multiplier, this choice sets the level of the published arm without changing its kinetics, and the response to warming, computed within each draw, does not depend on it.

1.16 Predictive evaluation and residual analysis

The fit of each model to the observations is evaluated with the transient calibration: for 100 posterior draws, the stock is simulated at each date and on each plot where soil carbon was measured, both on the calibration plots and on the held-out plots, which did not enter the calibration. Skill is reported, separately for the two sets, as the coefficient of determination between the observed stocks and the mean of the simulated ones, and as the root mean squared error between the observed stocks and the median of the simulated ones. Residuals are the logarithm of the ratio of the observed to the predicted stock, the scale of the error model; a residual of 0.1 corresponds to about 10 per cent more carbon in the soil than predicted. The residuals of any two models correlate at 0.99 or more, and the detailed analyses use one value per observation, the mean residual of Yasso07 and Yasso15.

These residuals were analysed in three ways. First, a random forest was fitted to the residuals of each model and evaluated out of bag, with about sixty site variables (climate, soil texture and chemistry, litter composition, the stand in 1985), to measure how much of the residual can be predicted and which variables lead; the ranking was checked with several hundred further variables from the four inventories, soil chemistry, topography, nitrogen deposition, management records and the understorey cover of 2023, leaving out variables that measure soil carbon itself.

Second, the mean residual was regressed on the leading predictor of the random forest, the basal area of the stand in 1985, together with the forest site type, the texture of the mineral soil recorded in 1985 and the volume share of stones.

Third, the residual of each observation was related to the time since the last regeneration cut (a felling that removes the stand) before sampling, dated from the inventories of 1985, 1990, 1995 and 2021–2023 and from the forest use declarations filed with the Finnish Forest Centre from 1997, or, where no cut was recorded, the age of the stand in 1985 plus the years since; the comparison with the basal area of 1985 used a regression with a separate intercept for each plot.

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